100 Days of Linux - Day 046

Title: Monitoring System Performance with top

Today's Goal
Learn how to monitor CPU, memory, and running processes in real time using the **top** command.

Why This Matters
Linux administrators, developers, and data engineers use **top** to identify resource-heavy programs, troubleshoot performance issues, and monitor long-running jobs.

Command of the Day: top

Common Syntax:
top

Useful Keyboard Shortcuts (inside top):
P = Sort by CPU usage
M = Sort by memory usage
1 = Show CPU cores (if supported)
k = Kill a process (requires PID)
q = Quit top

Important Fields:
PID, USER, %CPU, %MEM, TIME+, COMMAND

Beginner Lesson
Unlike **ps**, which shows a snapshot, **top** updates continuously. You can watch processes change in real time and identify programs using the most CPU or memory.

Practice Questions

1. Run the top command and observe the live process list for at least one minute.
2. Sort the process list by CPU usage using the correct keyboard shortcut.
3. Sort the process list by memory usage using the correct keyboard shortcut.
4. Identify the PID of the process currently using the most CPU, then quit top.
5. Display your current working directory after completing today's tasks.

Hints

Remember that keyboard shortcuts are pressed while top is running, not at the shell prompt.

Mini Challenge

Open another terminal and run 'yes > /dev/null'. In the first terminal, use top to find the high-CPU process, note its PID, then stop it using Ctrl+C in the other terminal.

Common Mistakes
Typing shortcut keys after exiting top. They only work inside the interactive top window.

Did You Know?
The top display refreshes automatically every few seconds, giving you a live view of system activity.

Pro Tip
Press q instead of closing the terminal window to exit top cleanly.

Answer Key (Instructor Only)
Q1: top
Q2: Press P
Q3: Press M
Q4: Note the PID shown at the top of the process list, then press q
Q5: pwd

Homework
Run top while opening and closing applications or terminal commands. Observe how CPU and memory usage changes.

Next Day Preview
Tomorrow you'll learn htop, a more user-friendly process viewer with color, mouse support, and easier navigation.